

Winning Strategy - Normal Play Nim

Normal Play = Person who takes last object loses

- If not nim sum of 0
 - Finish every move with nim sum of 0
 - To find out which move to make, let X be the nim-sum of all the heap sizes. Find a heap where the nim-sum of X and heap-size is less than the heap-size; the winning strategy is to play in such a heap, reducing that heap to the nim-sum of its original size with X .
 - If no such heap exists ...
- If nim sum 0, then person after will lose unless they make a mistake
- If there are only two heaps left, the strategy is to reduce the number of objects in the bigger heap to make the heaps equal. After that, no matter what move the opponent makes, the player can make the same move on the other heap, guaranteeing that they take the last object.

Nim Sum - xor all the sizes of the heaps represented by binary

- Then xor against all the individual heap sizes, and take from the pile in which the xor is less than the actual size of the heap
- Remove the nim sum of all heaps from that pile chosen

Winning Strategy - Misere Nim

- Only difference between misere and normal play optimal strategies in case of heaps of size 1
 - In misere leave an odd number of heaps of size one
- Same to keep nim sum of 0 until reaching 2 heaps with one heap size being exactly equal to one
 - Take the entire other heap, leaving the other player with the heap of 1